

Rishav Gupta

Ph.D. Student, Department of Computer Science and Electrical Engineering
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RESEARCH INTERESTS

Mobile health sensing and mobile systems, with an emphasis on acoustic and ultrasonic sensing, digital signal processing, and machine learning for cuff-free cardiovascular monitoring on commodity smartphones.

EDUCATION

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| University of Maryland, Baltimore County
<i>Ph.D. in Computer Science</i>
• Advisor: Dr. Dong Li , Future Sensing and Interaction (FSI) Lab . | Jan. 2026 – Present
Baltimore, MD |
| University of Maryland, Baltimore County
<i>M.S. in Computer Science</i>
• GPA: 4.00/4.00. Coursework: Design and Analysis of Algorithms, Machine Learning, Neural Networks. | Jan. 2025 – Dec. 2025
Baltimore, MD |
| Manipal Institute of Technology
<i>B.Tech. in Information Technology</i>
• Ranked 8th out of 132 students in the graduating class. | Jun. 2015 – May 2019
Manipal, India |

PUBLICATIONS

- [1] **Rishav Gupta**, Alyssa N. Maguina, Sithumina Weeraratna, Janelle P. Clark, and Dong Li. “WiP: Smartphone-based Acoustic Tissue Characterization for Personalized Body-contact Systems.” In *Proceedings of the ACM International Conference on Mobile Computing and Networking (MobiCom '26)*, Work-in-Progress Track. ACM, 2026. To appear.
- [2] **Rishav Gupta**, Shirui Cao, Jie Xiong, Sunghoon Ivan Lee, Jeremy Gummeson, and Dong Li. “Demo: Phonotonos: Through-Skin Ultrasonic Blood Flow Sensing Using Smartphones.” In *Proceedings of the ACM International Conference on Mobile Computing and Networking (MobiCom '26)*, Demo Track. ACM, 2026. To appear.
- [3] Renzhi Hao, Shirui Cao, Yizhu Wen, **Rishav Gupta**, Dharani Nadendla, Mehran Kafai, Hanqing Guo, and Dong Li. “Ultrasound Watermark: Real-time Acoustic Watermarking against Voice Scams on Smartphones.” In *Proceedings of the 9th ACM/IEEE Workshop on Security and Privacy in Edge Computing (EdgeSP '26)*, co-located with ACM/IEEE SEC. IEEE, 2026. To appear.
- [4] Shirui Cao, Jie Xiong, **Rishav Gupta**, Sunghoon Ivan Lee, Jeremy Gummeson, and Dong Li. “Phonotonos: Through-Skin Ultrasonic Blood Flow Sensing Using Smartphones.” In *Proceedings of the ACM International Conference on Mobile Systems, Applications, and Services (MobiSys '26)*, Emerging Ideas Track. ACM, 2026. [\[Paper\]](#) [\[Code\]](#)
- [5] Renzhi Hao, Shirui Cao, Yizhu Wen, **Rishav Gupta**, Dharani Nadendla, Mehran Kafai, Hanqing Guo, and Dong Li. “Ultrasound Watermark: Real-time Acoustic Watermarking against Voice Scams on Smartphones.” *Under review*, ACM MobiCom 2027. Extended version of [3].

HONORS AND AWARDS

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| Computing for Social Good Special Award , COEIT Research Day, UMBC
• One of only two awards presented across 114 posters; conferred by Dean Jeanne M. van Briesen. | May 2026 |
| First Place , Maryland NEW VENTURE™ Program Pitch Night, bwtech@UMBC
• Awarded for <i>PocketCardio</i> , a smartphone-based cardiovascular screening venture. | May 2026 |
| Best Poster Award , CSEE Research Day, UMBC | Apr. 2026 |
| Best Poster Award , COEIT Research Day, UMBC | May 2025 |

INDUSTRY EXPERIENCE

Gojek (GoTo Group)

Feb. 2022 – Sep. 2024

Senior iOS Engineer

Bengaluru, India

- Reduced annual platform cost by \$3M by replacing a third-party analytics SDK (CleverTap) with an in-house, open-source tool at full feature parity.
- Cut development time by 20% across the engineering team by automating the bi-weekly CI/CD release pipeline in Python and Shell on GitLab.
- Doubled QA throughput with a real-time Event Visualizer for verifying analytics events in situ, and cut roughly 3 MB from the app binary through SwiftSyntax AST-based dead-code detection.

Nuclei

Jul. 2019 – Feb. 2022

Senior iOS Developer and Team Lead

Bengaluru, India

- Founding iOS developer of FedMobile (Federal Bank); implemented 60% of shipped features, including Payments, Investments, Loans, Onboarding, and Card Management.
- Promoted to Team Lead; led four iOS developers and closed a four-month delivery gap between the iOS and Android platforms.

Hera doo

Jun. – Jul. 2018

Android Developer Intern

Mostar, Bosnia

- Built an end-to-end Android sales-order application from scratch, increasing field sales by 25%.

SERVICE AND LEADERSHIP

Chair, Graduate Assistant Advisory Committee (GAAC), UMBC Graduate Student Association

2026 – 2027

HR Head, ISTE Students' Chapter, Manipal Institute of Technology

2017 – 2018

Co-Event Head (Acumen), TechTatva '17, Manipal Institute of Technology

Oct. 2017

Management Committee Member, ISTE Students' Chapter, Manipal Institute of Technology

2016 – 2017

TECHNICAL SKILLS

Languages Swift, Python, Java, Shell, SQL, HTML/CSS

Methods Acoustic and ultrasonic signal processing, physiological sensing, tissue biomechanics, embedded audio, machine learning

Tools Xcode, SwiftUI, Android Studio, Git, GitLab CI/CD, L^AT_EX

REFERENCES

Dr. Dong Li

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